

Where the network binds, dispatching becomes road scheduling

Deadlock-free task assignment and passage sequencing: pushing up the lower edge of the capacity interval (research plan)

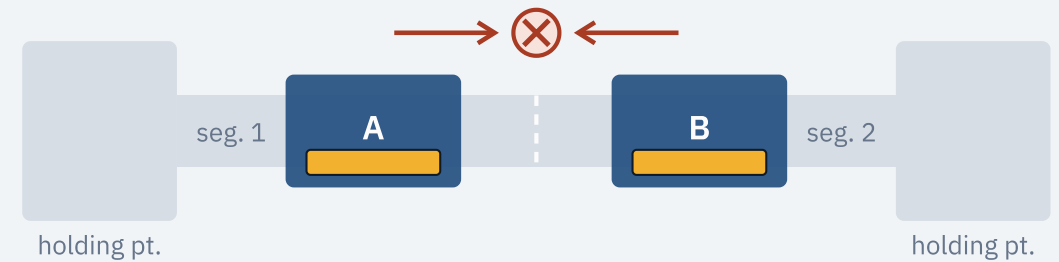
Deadlock-free task assignment and traffic orchestration for agents whose footprint grows when loaded (working title)

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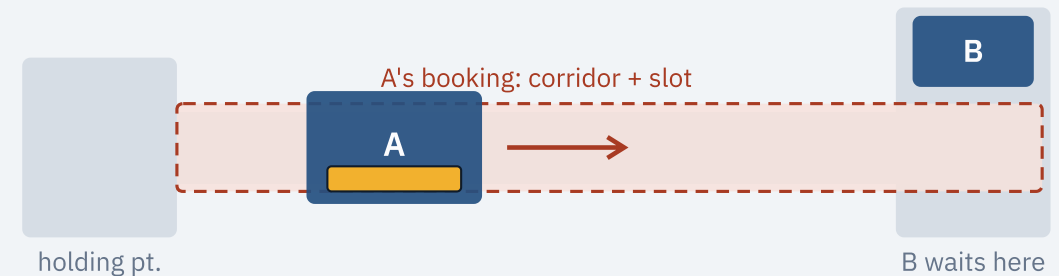
Grounded in chapter 3's reviewed results and this study's claims and hypotheses

Naive segment request: head-on in one lane



A holds 1, waits for 2; B holds 2, waits for 1

Safe-orchestration theorem: book to the next holding point



The slot is booked before release: no hold-and-wait cycle

This study in one minute

1

The question

Capacity is an interval, [best deadlock-free rule, network bound]. Okpo's main scenario gives [627, 1,720] a day: the lower edge is whole-route reservation, only 36% of the bound.

2

Starting point

Safe release is built: 860,291 decisions match a naive reduction, and it never deadlocks; but about 2/3 of its waits start from a criterion refusal, and from $K = 20$ on it stays below reservation.

3

The approach

The first rule comparison is running: a banker's-algorithm check plus a step-wise feasibility search, and claims cut at the next safe holding point (the safe-orchestration theorem). Then assignment and passage order are searched together inside that shell.

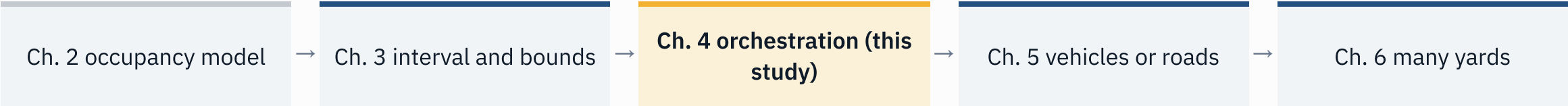
4

Outputs

A lower edge pushed towards field fleets, which under a 24-h day sit between free flow and reservation; an answer to the safe-orchestration cost hypothesis; a proof of the safe-orchestration theorem; a chapter 4 journal paper.

Where the network binds, how much of the bound is achieved is decided by traffic orchestration.

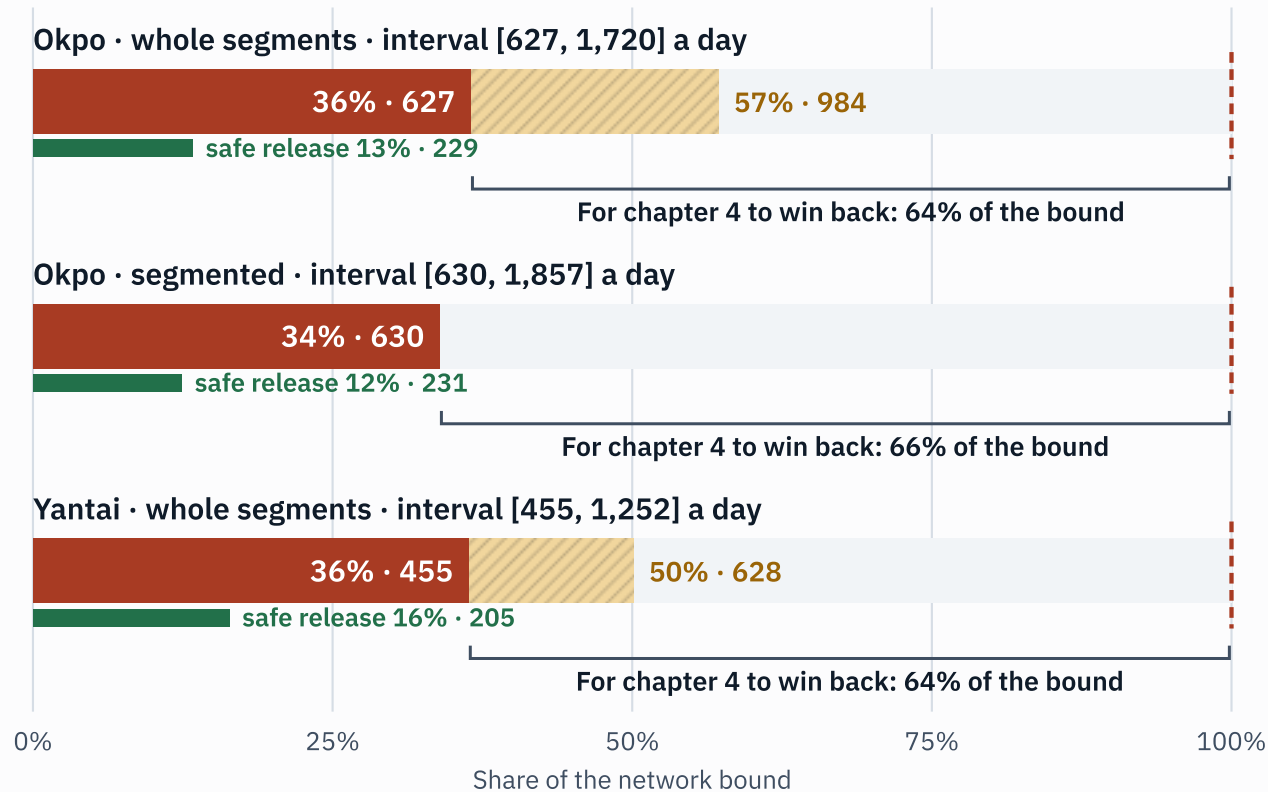
Its place in the thesis: chapter 3 gives the capacity interval, chapter 4 pushes its lower edge up



ITEM	WORDING IN THIS STUDY	HOW CHAPTER 4 ANSWERS
Two ceilings: interval	Achievable capacity lies in the interval [best deadlock-free rule, network bound]	Today's lower edge is whole-route reservation (safe release, also tested, is more conservative); the rules chapter 4 delivers take over, starting with the first rule comparison
From dispatching to traffic ordering	Where the network binds, how much of the bound is achieved is decided by traffic orchestration: release order, waiting points, reservation mode, block granularity. Naive dispatching deadlocks	All four levers become decision variables; deadlocks are counted only under no-teleport rules
Exchange rate: boundary	When the plateau lies far below the bound, change the orchestration first. This is the boundary between traffic ordering and the exchange rate	Okpo's main scenario: plateau / network bound = 0.36 (segmented 0.34), so orchestration comes first; erection mix 0.87 with whole segments, 0.53 segmented (page 5)
Safe-orchestration cost hypothesis	Naive dispatching deadlocks or nearly deadlocks at peaks; a provably safe policy loses only a little throughput	First reading on page 10 (K = 30: reservation 61%, safe release 46%); the rule comparison re-checks it
Safe-orchestration theorem	Safe orchestration: reserving to the next safe holding point never deadlocks, with a loss bound relative to the upper bound	Status “to do (chapter 4)”; being tested in the rule comparison (running); proof route on page 9

Sources: this study's claims and hypotheses, Chinese wording quoted verbatim and translated here; readings from simulation in this study (pages 4, 5, 10); chapter links follow the structure of the thesis.

The capacity interval [best deadlock-free rule, network bound]: on both yards the lower edge is whole-route reservation, about 36% of the bound



36% / 36%

Whole-route reservation plateau ÷ network bound (Okpo / Yantai, whole segments): the best deadlock-free rule, i.e. the lower edge

13% / 16%

Safe release ÷ network bound: also deadlock-free with no teleporting, but more conservative (page 8)

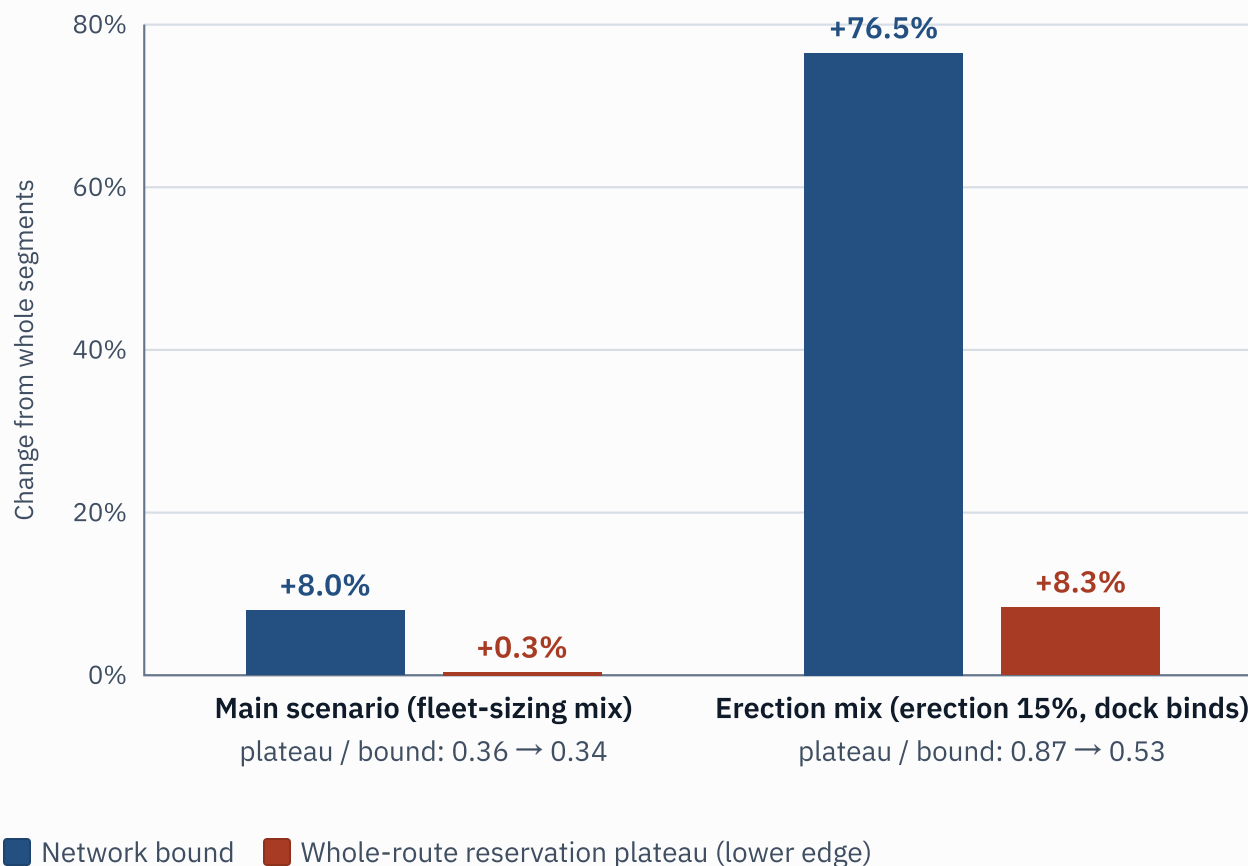
57% / 50%

Segment request (reference rule, teleport clearing) ÷ network bound: it relies on teleporting, so it cannot be the lower edge

- Whole-route reservation: best deadlock-free rule = lower edge
- Safe release
- Segment request (reference rule, teleport clearing)
- Network bound

Source: simulation in this study. Whole segments: Okpo (the fleet-sizing paper's task mix) and Yantai; safe release and segmented roads: the safe-release batches of both yards' main scenario. Conditions: saturated, 16-h day, 10 seeds; plateau = mean throughput over $K = 100-150$, network bound = mean bound over 10 seeds. Maps: the Okpo map (before floating docks and branch roads) and the Yantai map, with identical networks across batches; from the rule comparison on, Okpo moves to the map with floating docks and branch roads, while every reading in this briefing is on the former.

The rule, not the road, limits the main scenario: segmenting the dock roads lifts the bound by 8%, the lower edge by 0.3%



Ordering or exchange rate: two ready examples

Main scenario: the bound rises, the lower edge barely moves, so orchestration comes first. Erection mix (the dock binds): whole-segment reservation already reaches 87% of the bound, so road changes pay; segmented, the bound gains 76.5%, reservation 8.3%, and the ratio falls to 53%: the bottleneck moves to the rule.

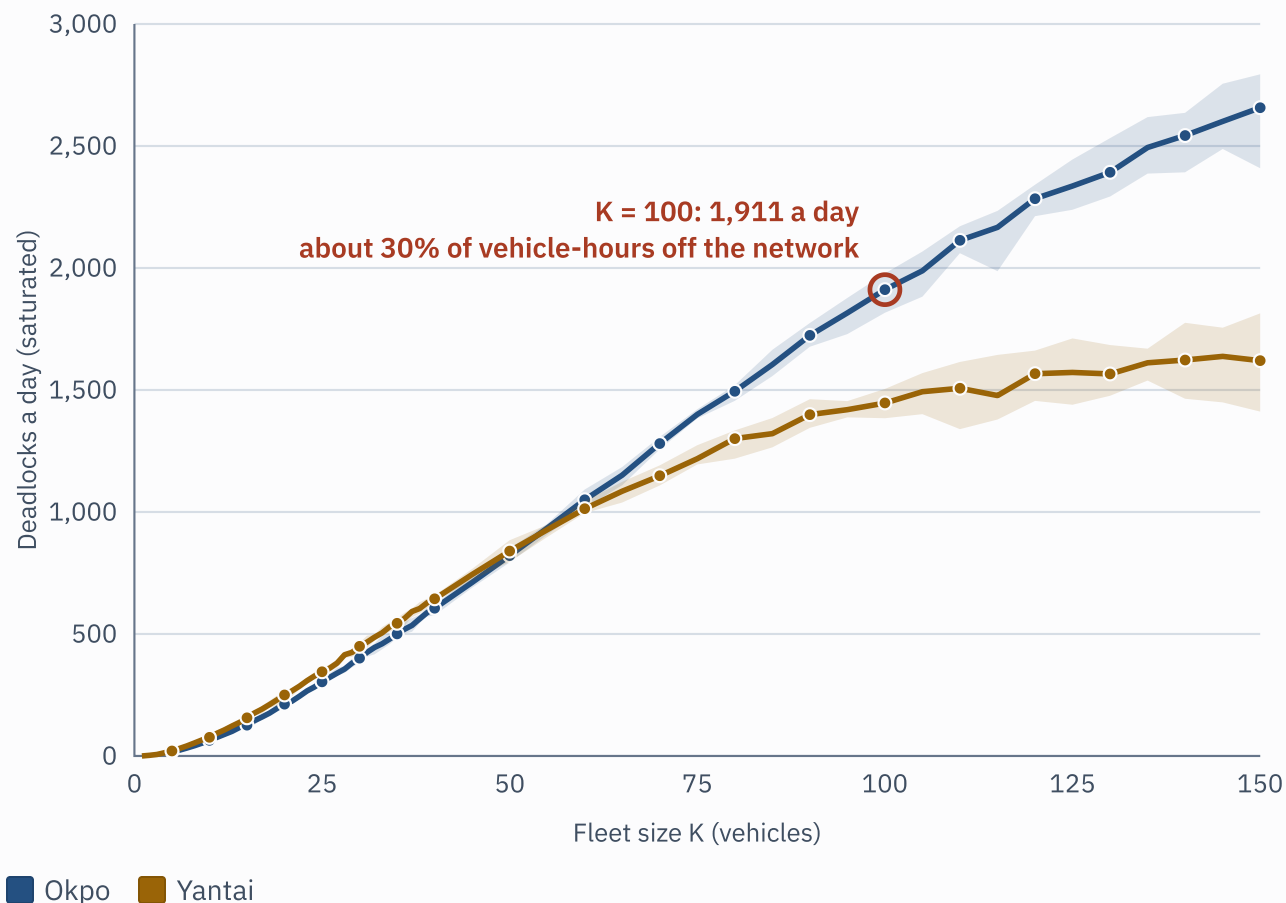
Small instances (checked by hand)

A 12 m single-lane corridor: reservation 19 a day, a hand-built feasible schedule 31.6, network bound 37.9. A crossroads: reservation 70, segment request (reference rule, teleport clearing) 105, network bound 187.

The cause is a rigid chain: reservation books empty run, loading, loaded run and unloading as one chain, so booked road idles during handling. Widening and passing bays get 10-seed numbers in the retrofit runs (3-seed probe: 12 m lifts the bound 5%, reservation 0.2–0.9%).

Source: simulation in this study. Main scenario: whole segments as on page 4, segmented from the main-scenario safe-release batch; erection mix: whole segments from the tightened-bound batches, segmented from the erection safe-release batch. Segmenting rule: roads longer than 200 m with roadside stops get block boundaries 25 m either side of each stop (a scenario setting; the retrofit simulations scan block length); on both yards the rule picks exactly the dock roads. Plateau = mean over $K = 100-150$; the network bound is the mean of 10 seeds. Small instances and the rigid chain: this study's tightened-bound analysis; the probe: a 3-seed trial (9 Oct 2026).

Naive segment request deadlocks; it survives only by “teleport” clearing, so it is a reference rule



26 / 83

Okpo deadlocks a day on a regular / peak day (97 / 194 tasks, the 4-vehicle subset of Yim 2008, for comparison only)

1,911 / 2,657

Deadlocks a day at saturation, K = 100 / 150

≈ 30%

Vehicle-hours teleported off the network at K = 100, at least (478 / 1,600)

Physical rules now give the reading: safe release never deadlocks; with no clearing the whole network gridlocks, faster as K grows (Okpo: about 4.3 h at K = 5, under 1 h from K = 40). Under the reference rule about 40–50% of loaded erection moves are teleported at least once on the way to the dock, so it serves as comparison only. Support from ports: a terminal network deadlocks from 30 vehicles without keep-clear hatching at junctions (Zhou et al. 2017).

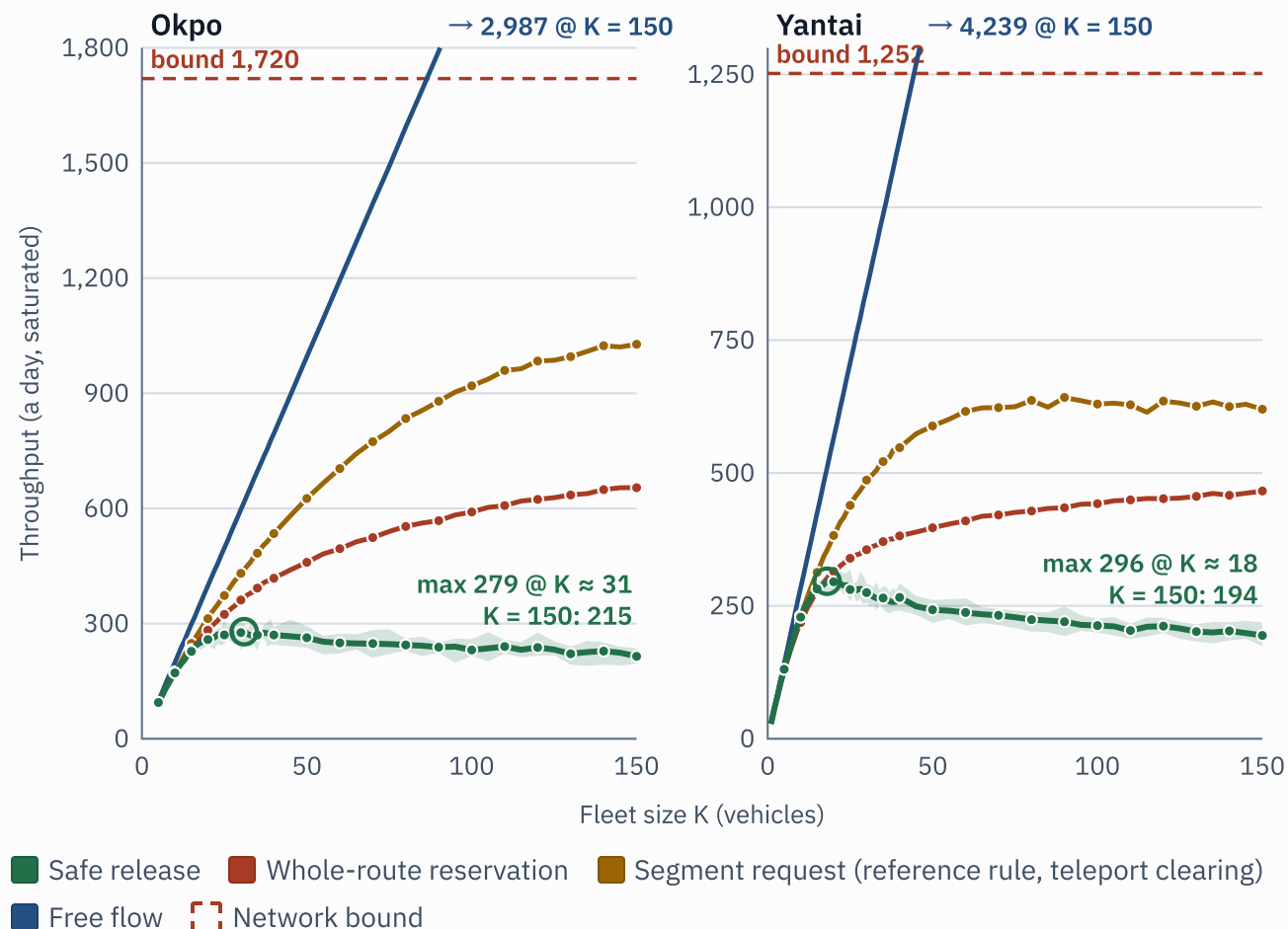
Source: simulation in this study, Okpo and Yantai (conditions as on page 4). Segment request (reference rule, teleport clearing): at each deadlock one vehicle releases all its resources, waits 900 s on open ground and requests again from where it was (no published value for 900 s; the flag experiment's setting). Deadlocks are means of 10 seeds, shading min–max; regular and peak are medians over the K grid. Vehicle-hours off the network = deadlocks × 900 s ÷ (K × 16 h), a lower bound. No clearing and teleports counted per move: this study's physical-rule runs.

Rules in use and in progress: reservation is today's lower edge, safe release is built, a rule comparison is trying two wider ones

RULE	HOW IT RELEASES	DEADLOCK	LIMITATION	USE IN CHAPTER 4	STATUS
Whole-route reservation	Books the whole route before a loaded departure	none	Rigid chain; plateau about 36% of the bound	Today's best deadlock-free rule: the lower edge	done
Segment request (reference rule, teleport clearing)	One segment at a time; at a deadlock one vehicle leaves for 900 s	yes	Outside the bound's proof; 40–50% of loaded erection moves teleported	Reference only	done
Segment request, no clearing	As above; deadlocked vehicles keep the road	yes	The whole network gridlocks, faster as K grows	Naive deadlocks (cost hypothesis, criterion 1)	done
Safe release	Grants a request only if the system stays in a safe state (a banker's-algorithm check)	none	Claims whole remaining tasks: 2/3 of waits start from refusals; below reservation for $K \geq 20$	Exact, 1.1–2.4× run time; chapter 4's start	done
Segmented blocking + direction locks	Long roads cut at their stops; one lane: follow one way, the other waits	no head-on	U-turns at stops bypass the lock (under 1% of deadlocks); fixed before the comparison	Block-granularity lever; Okpo's main scenario segmented	done
New: step-wise push; holding-point booking	Grant if the banker's check or a push succeeds; booking: depart only with all road to the next safe holding point	none (theorem to prove)	Holding-point placement and capacity set the loss	Wider, just as reliable; reports the refusal share	running

Sources: simulation in this study (flag experiment, tightened bound, physical rules); the first rule comparison is running, no results yet; imagery check: sites next to Okpo's dock-1 road stay reachable while a vehicle works at its dock edge, so the main scenario is segmented at the stops, whole segments as comparison (decided 9 Oct 2026); theorem wording as in this study.

Chapter 4's starting point: safe release is exact and never deadlocks, yet from $K = 20$ on it stays below whole-route reservation



860,291

decisions identical to a naive reduction without caching: 0 mismatches, 0 deadlocks

≈ 2/3

of safe-release waits begin with a criterion refusal while capacity is free (main scenario, all K : Okpo 61%–75%, Yantai 60%–70%)

279 → 215

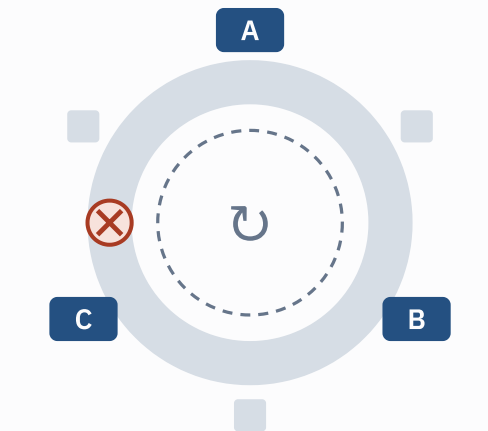
Okpo: max 279 @ $K \approx 31$, 215 at $K = 150$; Yantai 296 @ 18 → 194. The rise and fall belong to this criterion; the rule comparison re-checks them

Where it is conservative: the check assumes the others stand still and each vehicle finishes its whole remaining task in turn, so vehicles that would never meet can be judged unsafe (a six-entrance ring: three vehicles circling one way; the reference rule never deadlocks, safe release refuses once). Refused vehicles wait on the road. Run time: 1.1–2.4× the reference rule.

Source: simulation in this study (both yards' main scenario, whole segments); safe release and the other rules come from the batches of page 4. Saturated, 16 h, 10 seeds; the green shading is safe release's min–max; conditions as on page 4. Free flow ignores occupancy and is not drawn beyond the frame.

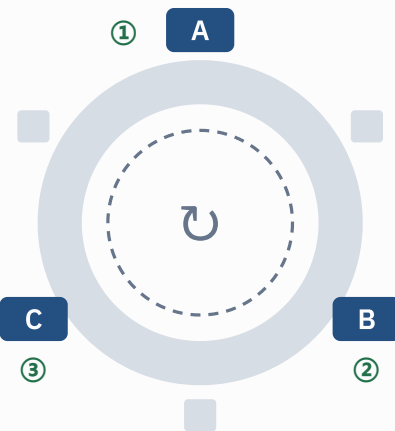
Running: chapter 4's first rule comparison tries two releases wider than the banker's check and just as reliable

(a) The banker's check (safe release)



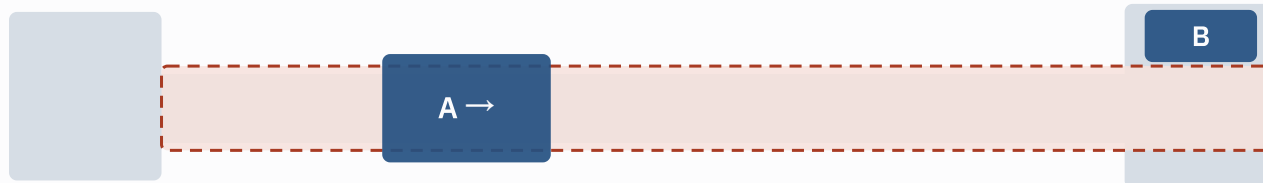
Each route crosses the next car's entrance
→ judged unsafe: C refused

(b) Step-wise feasibility push



Push one decision point at a time on a copy
→ all three finish: grant; the pushes are the plan

(c) Safe-orchestration theorem: claims cut at the next safe holding point

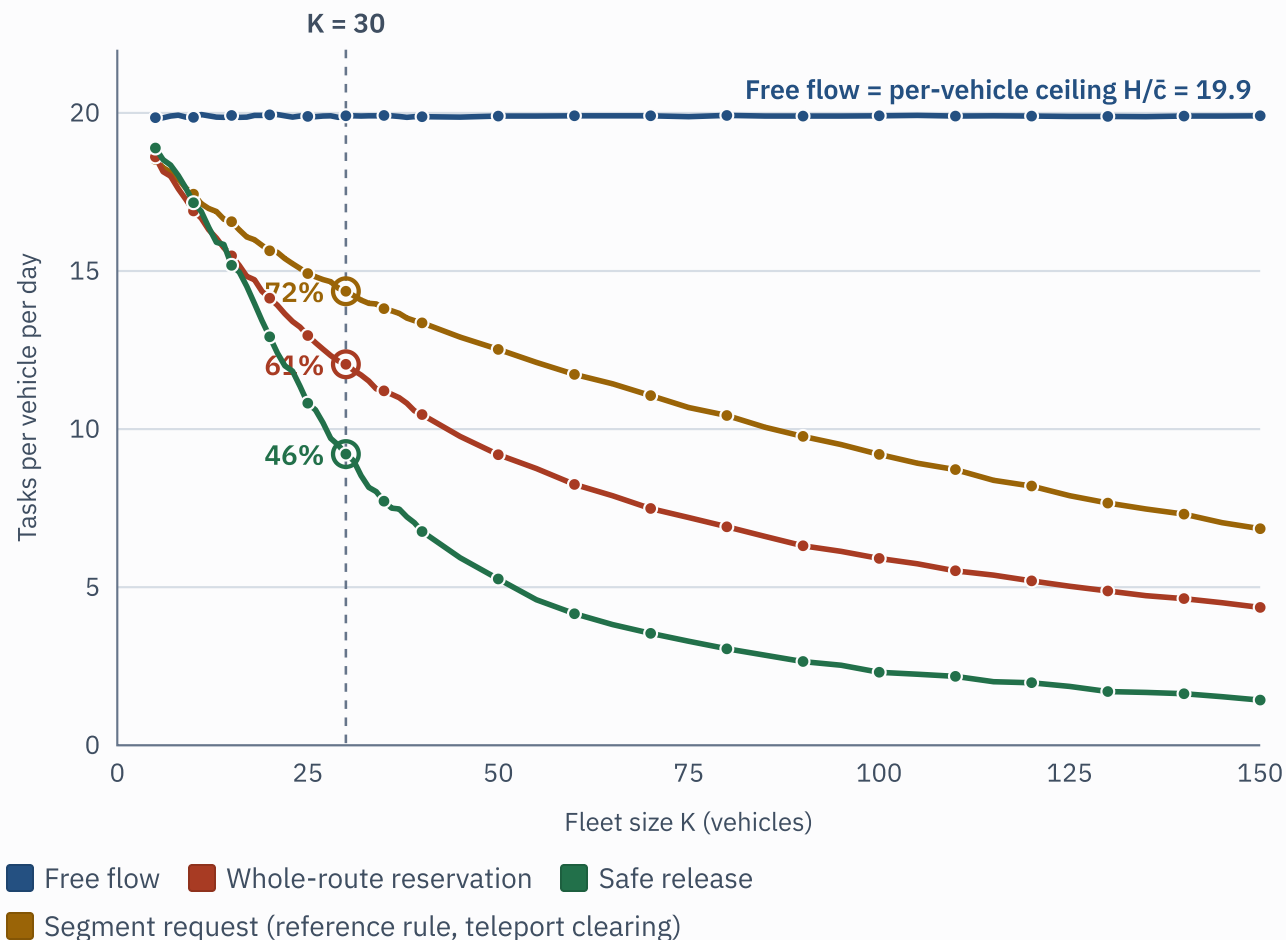


A books the corridor and a slot at the far end; B waits there: no hold-and-wait cycle

What the comparison does (started 10 Oct 2026; running, no results yet)

- **Two new rules:** grant if the banker's check holds or a step-wise push carries every vehicle to the end of its task (step-wise push: as reliable, never more conservative); or depart only with everything up to the next safe holding point (holding-point booking, the safe-orchestration theorem).
- **Readings:** each rule's share of waits caused by criterion refusals; re-checks of the lower edge, the decline hypothesis and the cost of safety (1 – best safe plateau / network bound).
- **Preparation:** new maps (Okpo with floating docks); U-turns at stops under the direction lock; reservation waits split into road and crane; resource names disambiguated.
- **Outputs and proof route:** the chapter 4 rule-comparison figure, the lower edge re-checked, and the theorem's sufficient condition with a draft proof (formal proof being written). It meets the network bound's conditions 1–3; max-plus gives its loss for a given order.

Cost of safety, first reading: at $K = 30$, reservation keeps 61% of free-flow efficiency, safe release 46%



61% / 46%

Per vehicle per day ÷ free-flow ceiling at $K = 30$: reservation / safe release; the reference rule (teleport clearing) reaches 72%

64% / 64%

The best safe rule's loss against the network bound (reservation; Okpo / Yantai), the hypothesis criterion; the teleporting reference rule still loses 43% / 50%

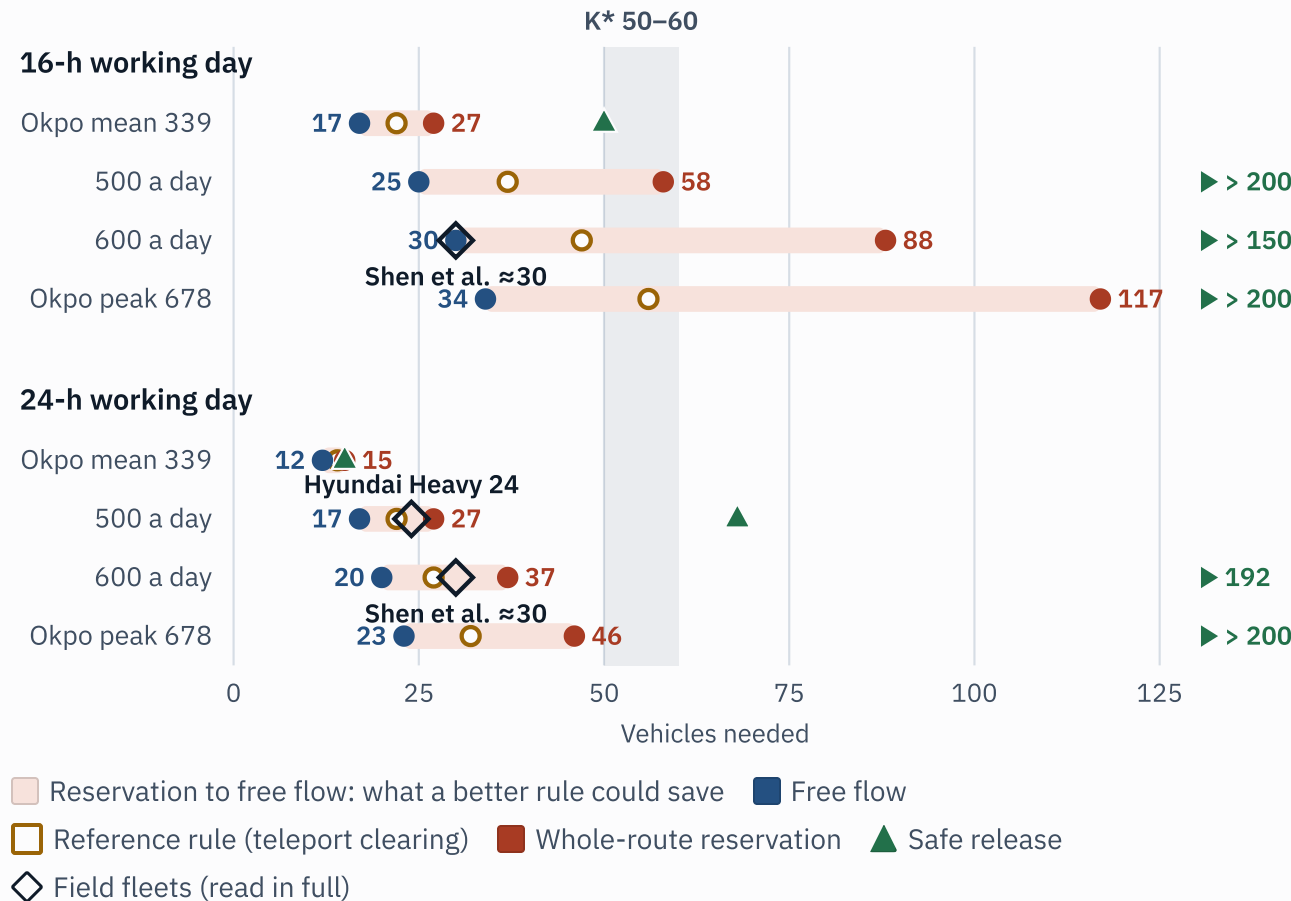
The safe-orchestration cost hypothesis

Criteria: deadlocks are counted under no-teleport rules; the cost of safety is the best safe rule's loss relative to the network bound. **If it holds:** safe orchestration is cheap. **If not:** a formula for the cost of safety and its conditions.

Efficiency is normalised by cycle time and meets the field once the anchor yards' haul distances are known; the loss against the bound is the hypothesis's criterion. The rule comparison re-checks both.

Source: simulation in this study (Okpo, 16 h, $K = 30$; main-scenario saturated curves as on page 8). Free-flow ceiling $H/\bar{c} = 960 \text{ min} \div \text{free-flow cycle of } 48.3 \text{ min}$ (empty 9.1, loading 10, loaded 19.1, unloading 10). Cost of safety = $1 - \text{best safe rule's plateau} / \text{network bound}$ (page 4), as read in the first rule comparison; hypothesis worded as in this study.

Whole-yard vehicles needed: what a better rule could save lies between reservation and free flow



88 → 30

Okpo at 600 a day, 16 h: whole-route reservation 88 vehicles, reference rule 47, free flow 30 (24 h: 37 / 27 / 20)

27 / 15

Okpo's own mean of 339 a day: 16 h reservation 27, free flow 17; 24 h 15 / 12, all below $K^* = 50-60$

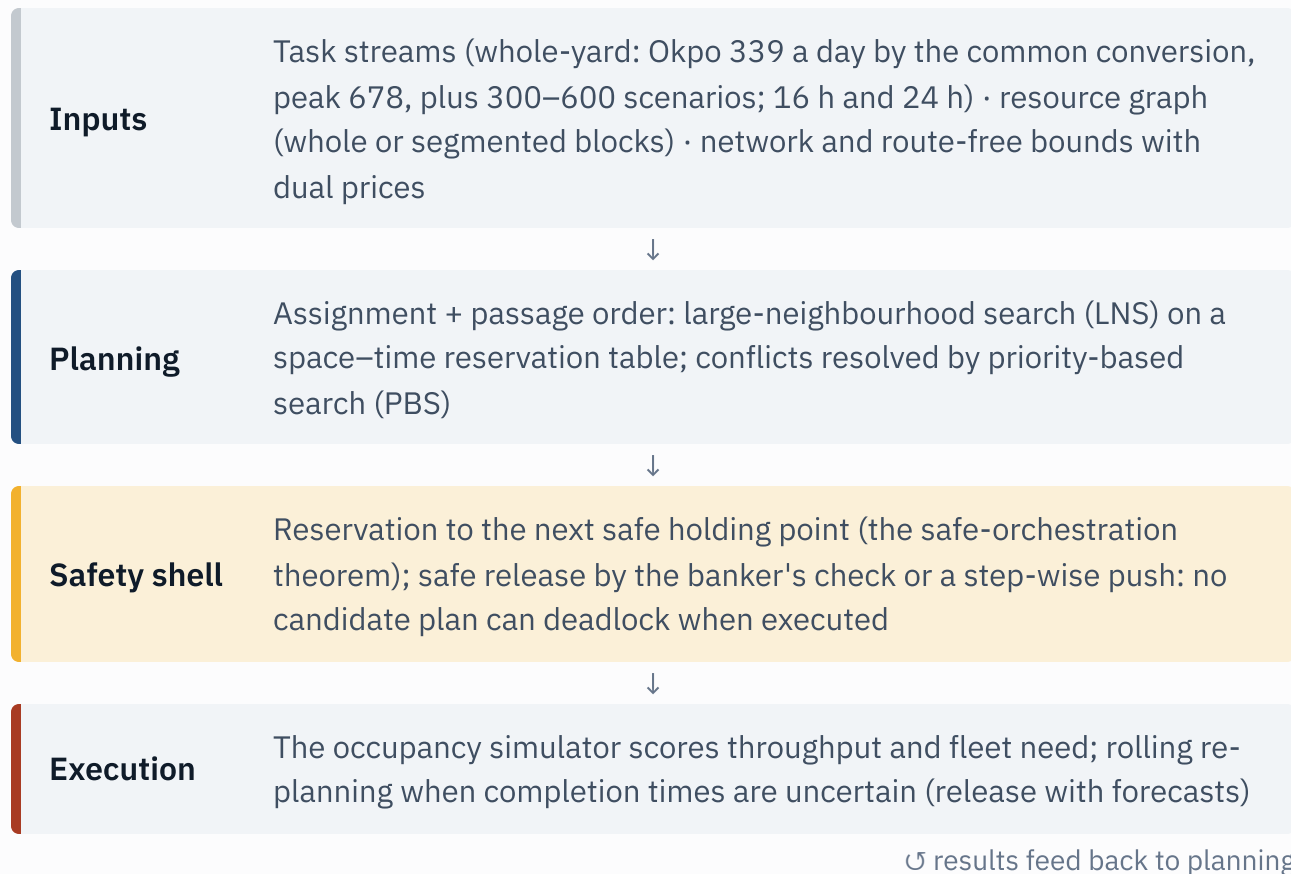
Where field fleets sit

Hyundai Heavy (about 500 moves a day, 24 h) runs 24 vehicles, between the model's 24-h free flow (17) and reservation (27); Shen et al. (about 600 a day, shift not stated) report about 30: between 20 and 37 at 24 h, while at 16 h only free flow (30) matches. At 24 h, real fleets sit between the two.

The main caliber (16 or 24 h, mean or peak) is set after the rule comparison; both are shown. At 16 h safe release misses 500 a day and above up to $K = 150$ or 200 (arrows).

Source: simulation in this study (Okpo whole segments: mean, 24-h and peak batches). Vehicles needed = smallest integer K whose mean throughput over 10 seeds reaches 99% of the daily volume (4 days, day 1 warm-up); 339 is Okpo's volume by the common conversion (external-anchor check), $678 = 339 \times 2$ the peak; $K^* = 50-60$ (network-bound test). Field fleets: 박창규 and 서준용 2006; Shen et al. 2018 (both read in full), taking a real fleet as equal to the vehicles needed (no maintenance or spare margin).

Methods: a safety shell around an integrated search over task assignment and passage order



Ground truth: CP-SAT exact model

One interval variable per move and resource, one no-overlap per segment and junction (cumulative where width is shared); optima on small instances.

Baselines

Whole-route reservation (today's best deadlock-free rule), safe release and the new rules, and “schedule in free flow, then repair conflicts”: they price integration and the shell separately.

Starting point: fleet-sizing code

ALNS and CP-SAT from the fleet-sizing paper; its CP-SAT already proves minimum fleets on 12–13-block batches.

[Fleet-sizing paper briefing →](#)

Optional: LLM-evolved rules

EOH, ReEvo, FunSearch (abstract level; no shipyard use found), compared with hand-written rules and optimisation.

Every component has a mature source; what is new is an agent that widens when loaded, long lifts and a capacity bound

COMPONENT	TRANSFER SOURCE AND THEORY	WHAT IS NEW IN ROAD-OCCUPYING SHIPYARD TRANSPORT
Exact model	Job shop with blocking (Mascis & Pacciarelli 2002); OR-Tools CP-SAT	Width-shared resources; asymmetric loaded and empty occupancy; long holds at loading points
Integrated search	MAPF-LNS (Li et al. 2021); ALNS (Ropke & Pisinger 2006); fleet-sizing paper code	Assignment and passage order searched together; empty runs need road too; fleet size is part of the decision
Conflict resolution	Priority-based search, PBS (Ma et al. 2019); rolling-horizon prioritised planning (Zheng et al. 2026)	Agents whose footprint multiplies when loaded; priorities tied to the reservation mode (whole route, to a holding point)
Safety shell	The banker's algorithm (Habermann 1969); resource allocation systems (Reveliotis 2000, 2005); AGV reservation with consistent priorities (Kim et al. 2006)	Holding points come from the network and vehicle class; the cost of safety is measured against the network bound
Passing and blocking	Kiel Canal traffic with passing at sidings (Lübbecke et al. 2019, abstract level); railway block sections (UIC 406)	Block granularity as an orchestration variable, linked to the retrofit runs' block-length scan
Optional: rule evolution	EoH, ReEvo, FunSearch (abstract level)	Readable release rules, tested for transfer across yards (links to chapter 6)

Closest in the same field: Tao et al. 2017 (full text) simulate one yard with road capacity counted in vehicles to compare vehicle types, with no blocking-time formalism, rule comparison, deadlocks or bound. Closest in AGVs: Li et al. 2025 (preprint), deadlock-free attachable AGVs via Petri nets and ALNS, extended here to loads that widen the footprint and to losses measured against the network bound.

Sources: full-text comparison with the closest work (Tao 2017, Li 2025 and Zheng 2026, full text). DOIs checked against Crossref. Abstract level: Mascis 2002, Reveliotis 2000 and 2005, Kim 2006, Lübbecke 2019, EoH, ReEvo, FunSearch; metadata only, to be read: Habermann 1969, Ma 2019, Li 2021.

Experiments: each one feeds a figure or table of chapter 4

#	EXPERIMENT	OUTPUT (CHAPTER 4)	NEEDS	STATUS
E1	The deadlock-free lower edge: safe release and reservation × whole / segmented blocks, both yards' saturated curves; gridlock with the no-clearing switch	Fig. 4-1 starting curves and interval; Table 4-1 deadlocks under no-teleport rules (cost hypothesis, criterion 1)	physical rules	done
E2	The first rule comparison: step-wise push and holding-point booking against the banker's check and reservation; refusal share; re-check of lower edge, decline and cost of safety	Fig. 4-2 rule comparison; Table 4-2 refusal share and run time	physical rules; system bound	running
E3	Small-instance ground truth and loss split: CP-SAT optima for a corridor, a crossroads and the three-resource dock network; max-plus throughput of the booking rule for a given order	Table 4-3 gaps between optimum, rules and heuristics; Fig. 4-3 rule loss against bound slack	system bound; retrofits	planned
E4	Integrated search (LNS + PBS + shell) against the three baselines; Okpo 339 / 678 and 300–600 a day, 16 h and 24 h	Fig. 4-4 daily volume vs. fleet need; Table 4-4 cost of safety (cost hypothesis, criterion 2)	E2	planned
E5	Holding points and block granularity: spacing, holding capacity S, block length	Fig. 4-5 cost of safety against holding-point spacing and block length	retrofits; floating docks	queued after E2
E6	Rolling online: release under completion-time forecast error	Fig. 4-6 throughput against forecast error	E4	planned
E7	External check: vehicles needed and per-vehicle efficiency normalised by cycle time against field anchors (Yim 2008, Heo 2013, Shen 2018, 박창규 2006)	Table 4-5 external check (same basis as chapter 3)	physical rules; anchors	first reading

Next, a constructive pilot: once the rule comparison names the best safe rule, run CP-SAT and LNS on E3's small instances, tune parameters and produce the first gap table. Small gaps: scale to both yards. Large gaps: revise the neighbourhoods and the shell, then continue.

What holds, what is running, what is queued, and risks with mitigations

Holds (chapter 3, all reviewed)

Both interval ends, reservation at the bottom; safe release exact, deadlock-free, conservative; gridlock without clearing; segmenting lifts only the main bound; route-free bound (draft proof); whole-yard fleet need, field anchors

Running (since 10 Oct 2026)

Rule comparison: step-wise push and holding-point booking, refusal shares, re-checks of lower edge, decline, cost of safety. Then the network-bound hypothesis is settled: new numbers, two-layer wording, main caliber (16 / 24 h, mean / peak)

Queued (after the comparison)

Retrofits and the exchange-rate table: block-length scan, three-resource dock network. Floating docks, finite holding area S , $K_0(S)$. Chapter 4 prototype: CP-SAT, LNS, theorem proof

RISK	MITIGATION
Safe release is conservative (2/3 of waits from refusals); reservation sets the lower edge for now	The rule comparison adds the step-wise push and holding-point booking; then a space–time table (PBS, LNS) lifts the edge further
With detours, the network bound (fixed loaded routes) no longer holds	The route-free bound (done) is the upper edge: route slack on Okpo's main scenario +3.7% (bound gap +5.3%), Yantai 0
CP-SAT solves only small instances	Small instances give ground truth; at scale, LNS with its gap to both bounds reported
Field anchors differ in working day; Okpo's shifts are not public	Report 16 h and 24 h (at 24 h real fleets sit between free flow and reservation); the main caliber is set after the comparison
Okpo moves to the map with floating docks and branch roads	Before the runs, the comparison reports the network and curve changes first; existing runs keep their registered maps

Target journals (paper 2, chapter 4): only journals in the school's 2026 directory

First choice (B)

- *IEEE Transactions on Automation Science and Engineering* (class B)
Deadlock avoidance and scheduling in automated handling; theorem and algorithms together
- *Transportmetrica A: Transport Science* (class B)
Online dispatching and network orchestration, with rigorous models and strong baselines

Reach (A)

- *IEEE Transactions on Industrial Informatics* (class A)
If rolling online orchestration is linked to a digital twin and closed with real or high-fidelity data
- *IEEE Transactions on Robotics* (class A)
If written as multi-agent planning for agents that change shape when loaded, with provable safety

Alternatives (B)

- *Engineering Applications of Artificial Intelligence* (class B)
Engineering mechanism, intelligent method, decision gains
- *Expert Systems with Applications* (class B)
When rules are learned or evolved by language models
- *IEEE Robotics and Automation Letters* (class B)
A short paper on multi-vehicle coordination and safe release

Supplementary (C)

- *Computers & Industrial Engineering* (class C)
- *Autonomous Agents and Multi-Agent Systems* (class C)
- *Journal of Manufacturing Systems* (class C)

B first; A is a reach, each with its condition. The school requires at least one B-or-above journal paper; paper 2 joins the others in the set.

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Other: UIC (2013) *UIC Code 406: Capacity*, 2nd ed.; Google OR-Tools CP-SAT; KED Global (2024) on Hanwha Ocean transporters and signallers.

In short: from a reliable starting point, push the deadlock-free lower edge up

Grounds

The rule sets the interval's lower edge; the best deadlock-free rule is whole-route reservation at 36% of the network bound; safe release is exact and deadlock-free but more conservative.

Approach

A rule comparison first tries wider safe releases (step-wise push, holding-point booking); then assignment and passage order are searched together inside the shell; CP-SAT ground truth; rolling online.

Outputs

Chapter 4's rule comparison and deadlock-free lower edge, the cost of safety, a proof of the safe-orchestration theorem, and a paper (first choice IEEE T-ASE or Transportmetrica A, both B).

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Where the network binds, schedule the road rather than the vehicles: orchestration decides how much of the bound is achieved.

Thank you.